

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 9.176122846119895
rotor delay [s]: 0.189866078685
gimbal delay [s]: 0.034469256436
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 2.188431141778527e-16
common-scale FD null norm: 1.2119171636250048e-10

Dimensionless coordinate

log_mass_scale	0.600775551056
log_second_moment_mode_xx	-1.11800402708
log_second_moment_mode_yy	-9.74600873484
log_second_moment_mode_zz	8.61801508819
log_second_moment_mode_xy	-0.65189206315
log_second_moment_mode_xz	-0.788810038122
log_second_moment_mode_yz	0.538744081907
dimensionless_cog_displacement_x	-0.165900964527
dimensionless_cog_displacement_y	-0.215675225287
dimensionless_cog_displacement_z	0.240167074486
log_force_effectiveness_1	0.736124143279
log_force_effectiveness_2	0.31989298987
log_force_effectiveness_3	0.0319988321867
log_force_effectiveness_4	0.557206157338

Physical representative

mass [kg]: 4.288214621246425
CoG body [m]: [-0.03380884 -0.04135066 0.05552216]
force effectiveness: [2.08782769 1.37698041 1.0325163 1.74578822]
inertia [kg m²]:
[2.939082456, 0.4796070052, 1.864138169]
[0.4796070052, 4.038497128, -0.7078034199]
[1.864138169, -0.7078034199, 1.462919348]
principal moments [kg m²]: [0.01440542 4.20584754 4.22024598]

Per-bag raw residual-wrench RMS

failure_1_single: [30.80918272 31.03841208 22.93632193 428.3335512 554.20642719
225.44606506]